Assignment 2

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Introduction & Exploratory Analysis

The data comes from the 1967 NIMH Schizophrenia Collaborative Study, a placebo-controlled randomized clinical trial examining changes in psychotic symptoms among patients with schizophrenia. For this analysis, the three medications are combined into a single treatment group.

Table 1: Summary Statistics for IMPS79 Across Timepoints

Week	Mean	SD	Sample Size
0	5.37	0.87	434
1	4.57	1.23	426
2	4.18	1.83	14
3	4.02	1.44	374
4	3.07	1.70	11
5	3.22	1.72	9
6	3.31	1.48	335

Initial Thoughts

Generally, IMPS79 scores look to decrease over time, though not necessarily in a linear fashion. The largest changes in IMPS79 scores occur between the initial weeks, and the changes become smaller over time. In addition, variation between patients also increases over time, an expected observation of longitudinal data.

Initially, I have two thoughts on how we can model this data. Firstly, we can use a mixed effect model and analyze individual variation within these trends as well as the treatment and time interaction term. Second, we could also use a ordinal mixed effect model by turning IMPS79 into an ordinal outcome. The upside of this is the analysis is more interpretable, if the categories have actual meaning, rather than arbitrary. However, we are also losing a lot of information. For instance, IMPS79 values of 3, 3.5, and 4 would be grouped into one category, but clearly there is a distinction that can be made between these 3 values. In addition, the assumption of proportional odds, which may not be realistic.

AI Initial Thoughts

Prompt: *Comment on the values you obtained in 1a, and what might be suggested for a statistical modeling of these data.*

The mean IMPS79 score decreases over time, from 5.37 at baseline to 3.31 at week 6, suggesting that psychiatric symptoms generally improve over the course of the study. However, the decrease does not appear to be perfectly linear. The largest drop occurs early in the study, while the amount of change becomes smaller later on. The standard deviation also increases after baseline, suggesting that patients become more heterogeneous in their responses over time.

The sample sizes vary substantially across the seven weeks. In particular, weeks 2, 4, and 5 contain very few observed measurements compared with weeks 0, 1, 3, and 6. Therefore, the means at these timepoints are much less reliable and should be interpreted cautiously. The presence of missing observations also makes a longitudinal modeling approach preferable to simply analyzing each week separately.

AI's Revised Thoughts

Prompt: *Here is the course textbook, with an emphasis on chapters 1-5, 9-11. Comment on the values you obtained in 1a, and what might be suggested for a statistical modeling of these data.*

The mean IMPS79 score decreases substantially over time, from 5.37 at baseline to 3.31 at week 6, suggesting that patients' psychiatric symptoms generally improve over the course of the study. The decline does not appear completely linear: the mean falls relatively quickly early in the study and then appears to level off later.

The sample sizes are highly unbalanced across weeks. Weeks 0, 1, 3, and 6 contain most of the observations, whereas weeks 2, 4, and 5 contain very few. This pattern is largely due to the study design, since the primary scheduled assessments were at weeks 0, 1, 3, and 6, although there is also some attrition by week 6. Therefore, the means at weeks 2, 4, and 5 should be interpreted cautiously and should not be given the same weight as the better-observed time points.

Because IMPS79 is repeatedly measured on the same patients, the observations within a patient are correlated, so an ordinary linear regression assuming independent observations would not be appropriate. A mixed-effects regression model for the continuous IMPS79 outcome would be more suitable. A random intercept could account for differences in patients' baseline severity, and a random time slope could allow patients to improve at different rates. The apparent nonlinear pattern over time also suggests that time may need to be modeled nonlinearly. For example, a square-root transformation of week could capture a relatively rapid initial improvement followed by slower subsequent change; alternatively, polynomial or other flexible time trends could be considered. Treatment group and a treatment-by-time interaction could subsequently be included to determine whether the trajectories differ between patients receiving antipsychotic medication and placebo.

Comparison of Responses

Compared with the initial AI response, the revised response was more specific about the structure of the data and proposed concrete modeling choices. This was more consistent with my own initial thoughts, where I suggested different forms of mixed-effect models.

Mixed-Effect Models

A mixed-effects model is well suited to these data because it can use all available observations even when patients are missing measurements at some timepoints. This is especially useful here because the number of observations differs substantially across weeks. Under the usual assumption that the missingness is ignorable, such as missing at random, the model can accommodate this unbalanced longitudinal structure without discarding patients with incomplete follow-up.

Linear Mixed-Effect Model

For this analysis, treatment group is initially ignored. First, we fit a linear mixed-effects model allowing both the intercept and the effect of time to vary across patients. The model is

$$Y_{ij} = b_{0i} + b_{1i} \cdot \text{Week}_{ij} + \epsilon_{ij}, \qquad b_{0i} = \beta_0 + u_{0i}, \quad b_{1i} = \beta_1 + u_{1i}.$$

Here, β_0 represents the population-average IMPS79 score at baseline and β_1 represents the population-average linear change in IMPS79 per week.

Table 2: Fixed Effects for Linear Mixed-Effects Model

Fixed Effect	Estimate	SE	p-value
Intercept	5.125	0.044	< 0.001
Week	-0.330	0.014	< 0.001

Table 3: Random Effects for Linear Mixed-Effects Model

Random-Effect Component	Estimate
Patient intercept SD	0.658
Patient slope SD	0.200
Intercept-slope correlation	0.284
Residual SD	0.836

Non-linear Mixed-Effect Models

To examine whether the overall trend in IMPS79 over time is curvilinear rather than linear, I next fit a quadratic mixed-effects model and a square-root model. The quadratic model is

$$Y_{ij} = b_{0i} + b_{1i} \cdot \text{Week}_{ij} + \beta_2 \cdot \text{Week}_{ij}^2 + \epsilon_{ij}, \qquad b_{0i} = \beta_0 + u_{0i}, \quad b_{1i} = \beta_1 + u_{1i},$$

and the square-root model can be modeled as

$$Y_{ij} = b_{0i} + b_{1i}\sqrt{\text{Week}_{ij}} + \epsilon_{ij}, \qquad b_{0i} = \beta_0 + u_{0i}, \quad b_{1i} = \beta_1 + u_{1i}.$$

Table 4: Fixed Effects for Quadratic Mixed-Effects Model

Fixed Effect	Estimate	SE	p-value
Intercept	5.2745	0.0477	< 0.001
Week	-0.5819	0.0358	< 0.001
Week squared	0.0425	0.0056	< 0.001

Table 5: Fixed Effects for Square-Root Mixed-Effects Model

Fixed Effect	Estimate	SE	p-value
Intercept	5.3877	0.0436	< 0.001
Square root of week	-0.8278	0.0353	< 0.001

Table 6: Comparison of Alternative Time Trends

Model	Parameters	AIC	BIC	Log-Likelihood
Linear	6	4886.19	4918.47	-2437.09
Quadratic	7	4831.68	4869.34	-2408.84
Square-root time	6	4732.47	4764.75	-2360.24

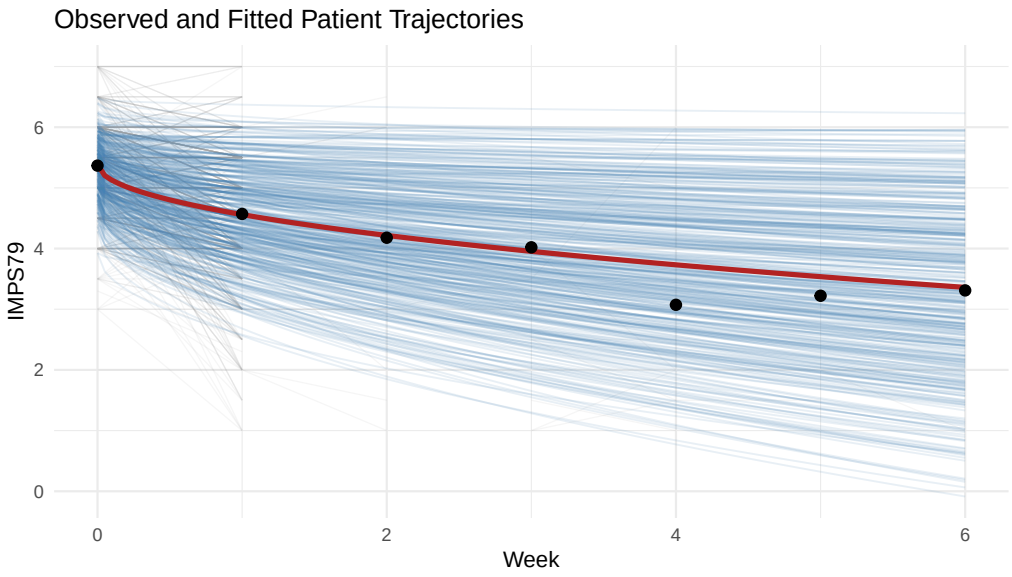
The results indicate that a simple linear time trend does not adequately describe the change in IMPS79 scores. The quadratic model improves on the linear model, with BIC decreasing from 4918.47 to 4869.34. However, the square-root time model provides an even better fit, with BIC = 4764.75. Because the square-root model achieves the best fit while requiring only a single transformed time coefficient, it was selected as the preferred specification for the longitudinal trend.

Under the square-root model, the estimated population-average intercept is 5.388 and the coefficient for square-root time is -0.828. However, the effect is not constant across calendar weeks: square-root time increases rapidly at first and then more slowly. This is consistent with the pattern observed in the sample means.

Table 7: Comparison of Random-Effect Structures for Square-Root Time Model

Model	AIC	BIC	Log-Likelihood
Random intercept	4968.97	4990.49	-2480.49
Random intercept + slope	4732.47	4764.75	-2360.24

The random-intercept and random-slope model fits substantially better than the random-intercept-only model. The likelihood-ratio test also strongly favors the random-slope model, $\chi^2(2) \approx 240.50$, $p < 0.001$.



The fitted trajectories show substantial patient-to-patient variation in both baseline IMPS79 levels and subsequent rates of change. However, most fitted trajectories follow the same general pattern of relatively rapid improvement early in the study followed by a more gradual decline at later weeks.

Treatment Effects

Since we've established that a square-root mixed effect model best fits the model above, this model is retained in this section, which explores the treatment effect of the drug. Let $\text{Drug}_i = 0$ denote placebo and $\text{Drug}_i = 1$ denote treatment.

Three models are considered. First, I modeled time but no interaction, which allows the treatment and placebo groups to differ in their overall levels while constraining them to follow the same time trend:

$$Y_{ij} = b_{0i} + b_{1i}\sqrt{\text{Week}_{ij}} + \beta_2 \text{Drug}_i + \epsilon_{ij}.$$

This provides a useful baseline model for assessing whether treatment is associated with an overall difference in symptom severity, without yet assuming that treatment changes the rate of improvement over time. Second, the full interaction model:

$$Y_{ij} = b_{0i} + b_{1i}\sqrt{\text{Week}_{ij}} + \beta_2 \text{Drug}_i + \beta_3 \left(\text{Drug}_i \times \sqrt{\text{Week}_{ij}}\right) + \epsilon_{ij}.$$

This is the most flexible of the three specifications because it allows the treatment and placebo groups to differ both at baseline and in their longitudinal trajectories. Finally, I fit a model without the treatment main effect:

$$Y_{ij} = b_{0i} + b_{1i}\sqrt{\text{Week}_{ij}} + \beta_3 \left(\text{Drug}_i \times \sqrt{\text{Week}_{ij}}\right) + \epsilon_{ij}.$$

This model constrains the two groups to have the same expected baseline IMPS79 score while still allowing their trajectories to diverge over time. This restriction is reasonable to consider in a randomized trial, where large systematic baseline differences between treatment groups are not expected, and it may provide a more parsimonious model if the treatment main effect is unnecessary. As above, for all three models,

$$b_{0i} = \beta_0 + u_{0i}, \qquad b_{1i} = \beta_1 + u_{1i},$$

where β_0 is the mean baseline IMPS79 score for the placebo group, β_1 is the square-root time trend for placebo, β_2 is the baseline difference between treatment and placebo, and β_3 is the difference in the time trend between treatment and placebo. Thus, β_3 is the primary parameter for determining whether treatment is associated with a different pattern of improvement over time.

Table 8: Fixed Effects: No Interaction Model

Fixed Effect	Estimate	SE	p-value
Intercept	5.596	0.083	< 0.001
Drug	-0.276	0.094	0.003
Square root of week	-0.828	0.035	< 0.001

Table 9: Fixed Effects: Full Interaction Model

Fixed Effect	Estimate	SE	p-value
Intercept	5.348	0.088	< 0.001
Drug	0.046	0.101	0.647
Square root of week	-0.336	0.068	< 0.001
Drug × square root of week	-0.641	0.078	< 0.001

Table 10: Fixed Effects: No Treatment Main Effect Model

Fixed Effect	Estimate	SE	p-value
Intercept	5.383	0.043	< 0.001
Square root of week	-0.346	0.064	< 0.001
Drug × square root of week	-0.627	0.072	< 0.001

Table 11: Comparison of Treatment Models

Model	Parameters	AIC	BIC	Log-Likelihood
Square-root time only	6	4732.47	4764.75	-2360.24
Treatment, no interaction	7	4727.01	4764.67	-2356.51
Full interaction	8	4665.00	4708.04	-2324.50
Interaction, no treatment main effect	7	4663.21	4700.87	-2324.60

The model comparisons provide strong evidence that treatment affects the trajectory of IMPS79 over time. Adding treatment alone to the square-root time model produces only a modest improvement in fit, whereas adding the treatment-by-time interaction results in a larger decrease in both AIC and BIC. This indicates that the primary treatment effect is not a constant difference between the treatment and placebo groups, but rather a difference in how their symptoms change over time.

The full interaction model and the model without the treatment main effect have similar AIC / BIC values. The treatment main effect in the full model is not statistically significant ($\hat{\beta}_2 = 0.046,\ p = 0.647$), while the treatment-by-square-root-time interaction is highly significant ($\hat{\beta}_3 = -0.641,\ p < 0.001$).

The preferred model is therefore the interaction model without the treatment main effect, because it has one fewer parameter and is therefore most parsimonious. In this model, the estimated square-root-time slope for treated patients is approximately $-0.346 - 0.627 = -0.973$, compared with -0.346 for placebo patients. This suggests that both groups improve over time, but patients receiving antipsychotic treatment improve substantially faster than patients receiving placebo.

Likelihood Ratio Tests

The likelihood-ratio tests further clarify the role of treatment in the model. First, adding a treatment main effect to the square-root time model significantly improves fit, $\chi^2(1) = 7.46,\ p = 0.006$. Adding the treatment-by-square-root-time interaction to the no-interaction model produces a substantial improvement in fit, $\chi^2(1) = 64.02,\ p < 0.001$. Finally, once the treatment-by-time interaction is included, adding the treatment main effect does not significantly improve model fit, $\chi^2(1) = 0.20,\ p = 0.655$. This supports the more parsimonious model containing square-root time and the treatment-by-time interaction, but no treatment main effect.

Table 12: Observed vs. Fitted Means: Placebo

Week	Observed	Fitted	Difference	n
0	5.35	5.38	-0.03	107
1	4.99	5.04	-0.05	105
2	5.80	4.89	0.91	5
3	4.74	4.78	-0.04	87
4	5.50	4.69	0.81	2
5	4.25	4.61	-0.36	2
6	4.25	4.53	-0.29	70

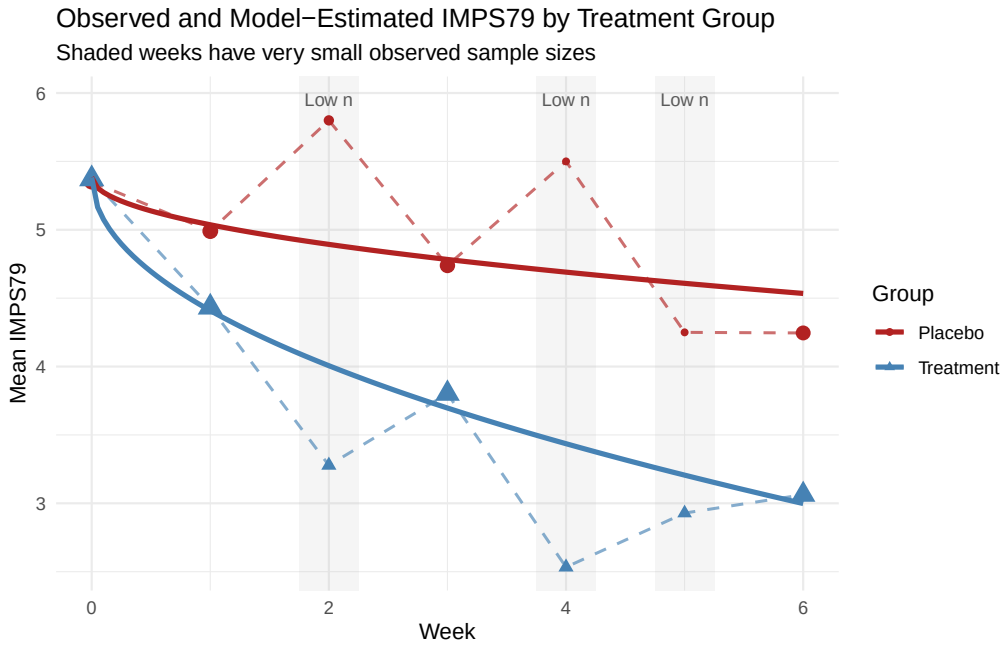
Table 13: Observed vs. Fitted Means: Treatment

Week	Observed	Fitted	Difference	n
0	5.37	5.38	-0.01	327
1	4.43	4.41	0.02	321
2	3.28	4.01	-0.73	9
3	3.80	3.70	0.11	287
4	2.53	3.44	-0.90	9
5	2.93	3.21	-0.28	7
6	3.06	3.00	0.06	265

The fitted means generally follow the observed treatment-specific patterns over time. The treatment and placebo groups begin at similar IMPS79 levels, consistent with the lack of evidence for a treatment main effect at baseline. After baseline, however, the treatment group declines more rapidly than the placebo group, consistent with the significant treatment-by-square-root-time interaction in the final model.

Thus, the final treatment model is

$$Y_{ij} = b_{0i} + b_{1i}\sqrt{\text{Week}_{ij}} + \beta_3 \left(\text{Drug}_i \sqrt{\text{Week}_{ij}}\right) + \epsilon_{ij}, \qquad b_{0i} = \beta_0 + u_{0i}, \quad b_{1i} = \beta_1 + u_{1i}.$$



Conclusion

Overall, IMPS79 scores decrease over the six-week study period, but the change is nonlinear and varies substantially across patients. A mixed-effects model using square-root time with patient-specific random intercepts and slopes provided the best representation of the longitudinal trend. After incorporating treatment, the preferred model included a treatment-by-square-root-time interaction but no separate treatment main effect. The treatment and placebo groups therefore begin at similar expected symptom levels, but patients receiving antipsychotic medication show a substantially faster decline in IMPS79 over time.

Code

The complete R code used for this analysis can be found here:

GitHub Repository